THE OPEN SUBFUNCTOR F IS REPRESENTABLE BY AN OPEN OF HILB^G

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1213-1219 of sources/stacks-pic/pinned-original-source.tex; proof span is lines 1221-1314, the complete author proof (92 body lines).

Editorial note. Navigation stubs: 9 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 2 of queue_pic.json (pb=3806, stmt 1213-1219, proof 1221-1314).

Editorial note. stacks-pic wave-1 (pi01): fresh adjudication accept (2026-10-14); pool per standing queue hypercovering -> pic (SMC reassigned to Zhen per Chief 08:5x); pic.tex 27 labels/23 with proof judged 21 accept / 2 reject / 4 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed. Named omission: Author omits the verification that the assignment L |-> m is an inverse transformation, and leaves 'a nonzero section defines an effective Cartier divisor' as clear - named mechanical checks; the W construction and the divisor claim are in-span.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 001P: sources/stacks-pic/pinned-original-source.tex, source lines 1314–1314. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 062Y: sources/stacks-pic/pinned-original-source.tex, source lines 1314–1314. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0B9V: sources/stacks-pic/pinned-original-source.tex, source lines 1187–1195. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0B9D: sources/stacks-pic/pinned-original-source.tex, source lines 625–634. Cited by the retained statement or proof but not part of the

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collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0B91: sources/stacks-pic/pinned-original-source.tex, source lines 1314–1314. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0B9S: sources/stacks-pic/pinned-original-source.tex, source lines 1314–1314. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0B9I: sources/stacks-pic/pinned-original-source.tex, source lines 849–867. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0B9J: sources/stacks-pic/pinned-original-source.tex, source lines 912–931. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0AYR: sources/stacks-pic/pinned-original-source.tex, source lines 1314–1314. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let k be a field. Let X be a smooth projective curve of genus g over k with a k-rational point σ. The open subfunctor F defined in Lemma 0B9V is representable by an open subscheme of $\underline{\mathrm{Hilb}}^g_{X/k}$.*

Proof. In this proof unadorned products are over $\mathrm{Spec}(k)$. By Proposition 0B9I the scheme $H = \underline{\mathrm{Hilb}}^g_{X/k}$ exists. Consider the universal divisor $D_{univ} \subset H \times X$ and the associated invertible sheaf $\mathcal{O}(D_{univ})$, see Remark 0B9J. We adjust by tensoring with the pullback via $\sigma_H : H \to H \times X$ to get

$$\mathcal{L}_H = \mathcal{O}(D_{univ}) \otimes_{\mathcal{O}_{H\times X}} \mathrm{pr}_H^* \sigma_H^* \mathcal{O}(D_{univ})^{\otimes -1} \in \mathrm{Pic}_{X/k,\sigma}(H)$$

By the Yoneda lemma (Categories, Lemma 001P) the invertible sheaf $\mathcal{L}_H$ defines a natural transformation

$$h_H \longrightarrow \mathrm{Pic}_{X/k,\sigma}$$

Because F is an open subfuctor, there exists a maximal open $W \subset H$ such that $\mathcal{L}_H|_{W\times X}$ is in $F(W)$. Of course, this open is nothing else than the open subscheme constructed in Derived Categories of Schemes, Lemma 0B9S with $i = 0$ and $r = 1$ for the morphism $H \times X \to H$ and the sheaf $\mathcal{F} = \mathcal{O}(D_{univ})$. Applying the Yoneda lemma again we obtain a commutative diagram

$$\begin{array}{ccc} h_W & \longrightarrow & F \\ \downarrow & & \downarrow \\ h_H & \longrightarrow & \mathrm{Pic}_{X/k,\sigma} \end{array}$$

To finish the proof we will show that the top horizontal arrow is an isomorphism.

Let $\mathcal{L} \in F(T) \subset \mathrm{Pic}_{X/k,\sigma}(T)$. Let $\mathcal{N}$ be the invertible $\mathcal{O}_T$-module such that $Rf_{T,*}\mathcal{L} \cong \mathcal{N}[0]$. The adjunction map

$$f_T^*\mathcal{N} \longrightarrow \mathcal{L} \quad \text{corresponds to a section } s \text{ of } \quad \mathcal{L} \otimes f_T^*\mathcal{N}^{\otimes -1}$$

on X_T. Claim: The zero scheme of s is a relative effective Cartier divisor D on $(T \times X)/T$ finite locally free of degree g over T.

Let us finish the proof of the lemma admitting the claim. Namely, D defines a morphism $m : T \to H$ such that D is the pullback of D_{univ}. Then

$$(m \times \mathrm{id}_X)^* \mathcal{O}(D_{univ}) \cong \mathcal{O}_{T \times X}(D)$$

Since $\mathcal{O}_{T \times X}(D) \cong \mathcal{L} \otimes f_T^* \mathcal{N}^{\otimes -1}$ we conclude that $(m \times \mathrm{id}_X)^* \mathcal{L}_H$ and $\mathcal{L}$ differ by the pullback of an invertible module on T. This shows that $m : T \to H$ factors through the open $W \subset H$ above. Moreover, it follows that these invertible modules define, after adjusting by pullback via σ_T as above, the same element of $\mathrm{Pic}_{X/k,\sigma}(T)$. Chasing diagrams using Yoneda's lemma we see that $m \in h_W(T)$ maps to $\mathcal{L} \in F(T)$. We omit the verification that the rule $F(T) \to h_W(T)$, $\mathcal{L} \mapsto m$ defines an inverse of the transformation of functors above.

Proof of the claim. Since D is a locally principal closed subscheme of $T \times X$, it suffices to show that the fibres of D over T are effective Cartier divisors, see Lemma 0B9D and Divisors, Lemma 062Y. Because taking cohomology of $\mathcal{L}$ commutes with base change (Derived Categories of Schemes, Lemma 0B91) we reduce to $T = \mathrm{Spec}(K)$ where K/k is a field extension. Then $\mathcal{L}$ is an invertible sheaf on X_K with $H^0(X_K, \mathcal{L}) = K$ and $H^1(X_K, \mathcal{L}) = 0$. Thus

$$\deg(\mathcal{L}) = \chi(X_K, \mathcal{L}) - \chi(X_K, \mathcal{O}_{X_K}) = 1 - (1 - g) = g$$

See Varieties, Definition 0AYR. To finish the proof we have to show a nonzero section of $\mathcal{L}$ defines an effective Cartier divisor on X_K. This is clear. □